

RECOVERABLE ASSETS & INVENTORY RISK MANAGEMENT

GROUP 5



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Problem

1. Financial Risk Due to Inventory Obsolescence

- Excess inventory exceeding 12 months of consumption is classified as Recoverable Assets (RA)
- Magna Carta (MC) Dead Stock and Expired materials require financial write-downs
- Quantifying the total provision amount manually is inefficient and time-consuming

2. Lack of Integrated Data Pipeline

- Inventory, materials, purchase orders, production orders, sales orders, and supplier data are stored separately
- No automated process for data ingestion, cleaning, and transformation

3. Stockout & Order Fulfillment Risks

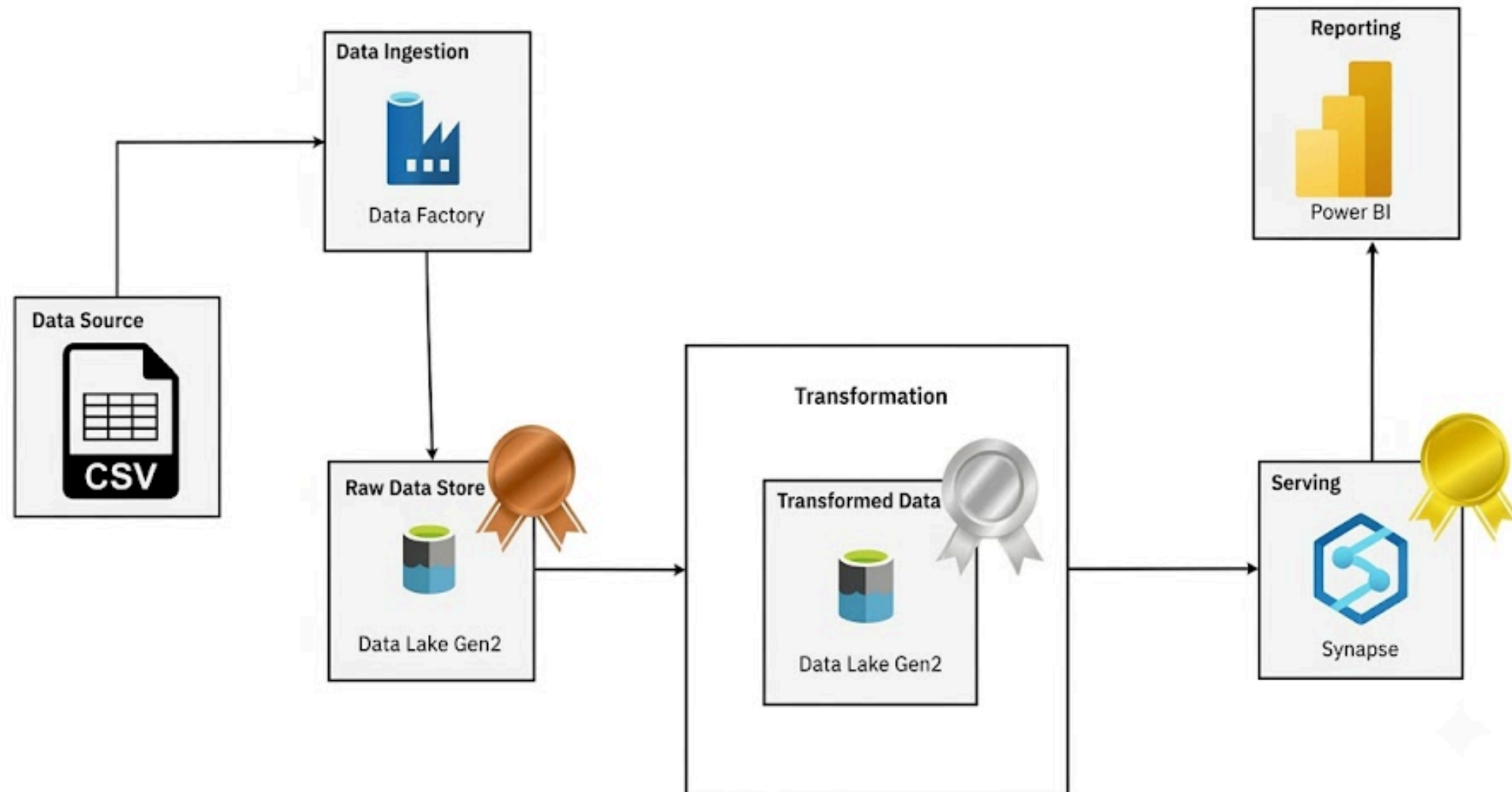
- Difficulty identifying materials at risk of stockout
- Limited visibility into affected customer sales orders
- May lead to delivery delays and customer dissatisfaction

Objectives

- To analyze six supply chain datasets to identify Recoverable Assets (RA), Magna Carta (MC), and stockout risks
- To design a three-layer data engineering architecture with Bronze, Silver, and Gold layers
- To implement an end-to-end data pipeline using Microsoft Azure for data ingestion, cleaning, and transformation
- To develop and evaluate an interactive Power BI dashboard to answer the defined business questions (Q1-Q7)



Data/System Architecture





Data Pipeline Workflow

BRONZE LAYER

SILVER LAYER

GOLD LAYER

INSIGHTS

RAW INGESTION



Pipeline Orchestration

Engineered 6 automated Azure Data Factory (ADF) pipelines to reliably extract operational CSVs (materials, inventory_snapshot, purchase_orders, production_orders, suppliers, sales_orders) into ADLS Gen2 landing zones.



Data Simulation

Expanded synthetic datasets to over 1,000 records each to stress-test the system, and injected the critical lot_expiry_date column necessary for Magna Carta calculations.

DATA TRANSFORMATION



PySpark Processing

Developed modular Synapse Spark notebooks to read raw data, standardize missing date formats, and execute staged data cleansing prior to business rule application.



Business Logic (RA/MC)

Programmed core algorithms to calculate 12-month rolling consumption (Recoverable Assets), and dynamically flag Magna Carta Dead Stock (>9 months zero usage) and Expired lots.



Financial Provisions

Automated the 100% financial write-down provision by multiplying MC-flagged on-hand quantities by unit costs, writing the enriched outputs as highly optimized Parquet files.

DATA SERVING



Star Schema Modeling

Architected the dimensional model by replacing natural keys with narrower integer surrogate keys, centering on fact_inventory_status connected to 4 specific dimension tables (dim_material, dim_supplier, dim_date, dim_sales_order).



Semantic Layer

Engineered the Synapse Serverless SQL semantic layer by creating dedicated views (vw_q1 to vw_q7) to pre-compute logic and answer the 7 core business questions natively in the cloud.



BI Integration

Structured the final data warehouse layout to eliminate complex recursive joins, ensuring high-performance DirectQuery or Import rendering.

INSIGHTS



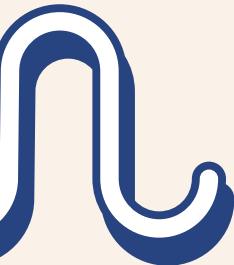
Interactive Dashboard

Designed a comprehensive Power BI interface utilizing cross-filtering slicers (by category, date, supplier, and risk type) to dynamically answer all 7 business questions.

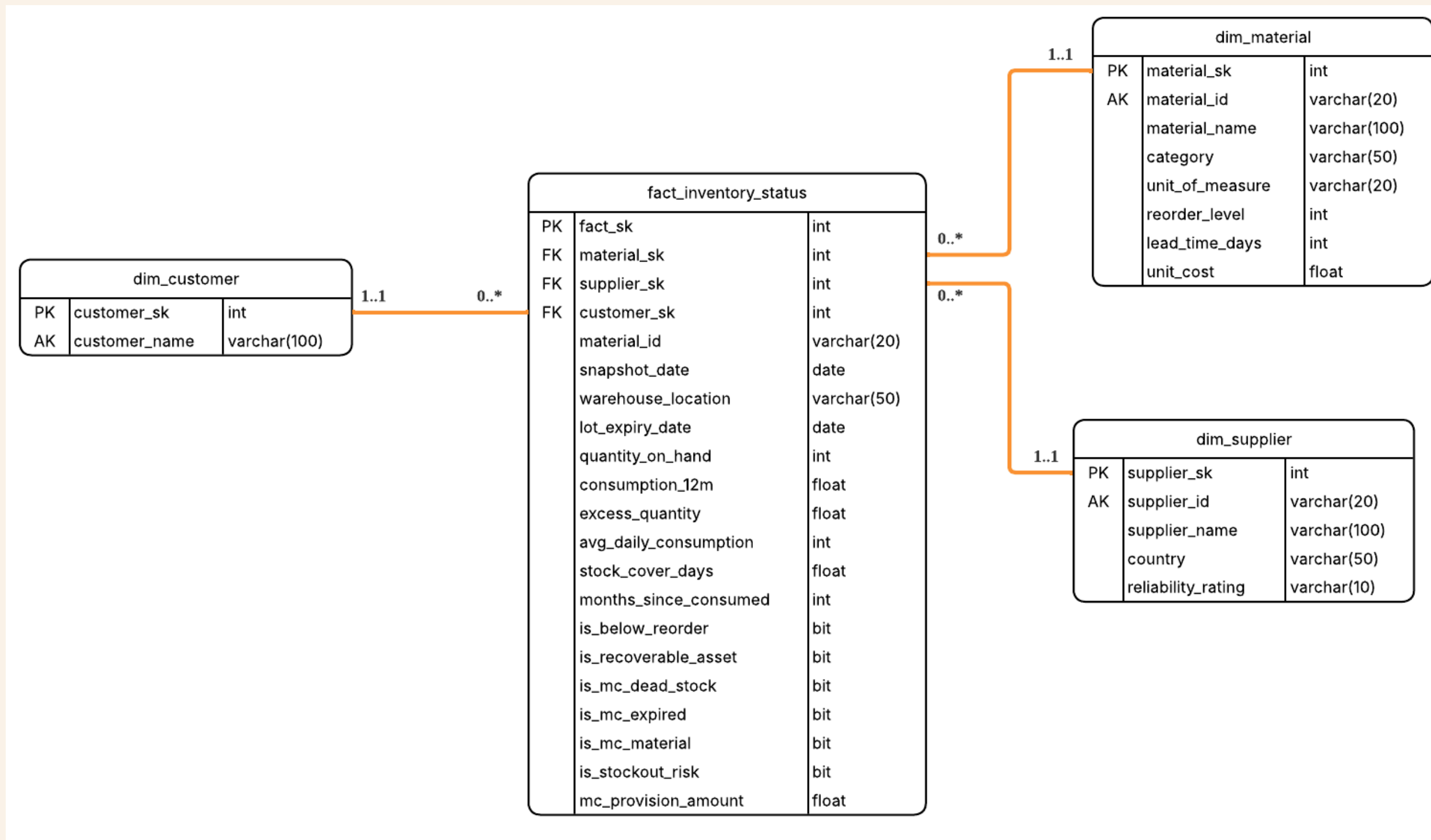


Stockout Alerts

Created targeted supply chain visualizations that cross-reference raw material shortages directly against open sales orders to trace exact customer delivery impacts.



Gold Layer Star Schema



Gold Tier Semantic Layer

vw_q1_below_reorder:

Isolates inventory that has dropped below safety limits to instantly trigger warehouse restocking alerts.

```
-- View 1: Q1 - Materials Below Reorder Level (Remains stable)
CREATE OR ALTER VIEW vw_q1_below_reorder AS
SELECT
    f.material_id,
    m.material_name,
    m.category,
    f.quantity_on_hand,
    m.reorder_level,
    f.snapshot_date,
    CASE WHEN f.quantity_on_hand < m.reorder_level THEN 1 ELSE 0 END AS is_below_reorder
FROM fact_inventory_status f
JOIN dim_material m ON f.material_id = m.material_id
WHERE f.quantity_on_hand < m.reorder_level;
GO
```

vw_q3_mc_dead_stock:

Isolates inventory stagnant for over 9 months into a dedicated ledger to prevent double-counting financial liabilities.

```
-- View 3: Q3 - MC Dead Stock (Keeps our strict hierarchy to prevent double-counting)
CREATE OR ALTER VIEW vw_q3_mc_dead_stock AS
SELECT
    f.material_id,
    m.material_name,
    m.category,
    f.quantity_on_hand,
    (f.quantity_on_hand * m.unit_cost) AS inventory_value,
    f.months_since_consumed,
    f.mc_provision_amount
FROM fact_inventory_status f
JOIN dim_material m ON f.material_id = m.material_id
WHERE f.is_mc_dead_stock = 1 AND f.is_mc_expired = 0;
GO
```

vw_q4_mc_expired:

Captures physically ruined inventory with dynamic batch expiry dates to guide warehouse disposal teams. Isolates

```
-- View 4: Q4 - MC Expired (NOW INCLUDES THE REAL EXPIRY DATE)
CREATE OR ALTER VIEW vw_q4_mc_expired AS
SELECT
    f.material_id,
    m.material_name,
    m.category,
    f.quantity_on_hand,
    f.lot_expiry_date,
    f.snapshot_date,
    f.is_mc_expired AS [is_expired_proof],
    f.mc_provision_amount AS [financial_loss_amount]
FROM fact_inventory_status f
JOIN dim_material m ON f.material_id = m.material_id
WHERE f.is_mc_expired = 1;
GO
```

vw_q5_recoverable_assets:

Tracks viable corporate inventory lines whose stock levels are safely backed by active annual consumption.

```
CREATE OR ALTER VIEW dbo.vw_q5_recoverable_assets AS
SELECT
    f.material_id,
    m.material_name,
    m.category,
    f.quantity_on_hand,
    (f.quantity_on_hand * m.unit_cost) AS inventory_value,
    f.consumption_12m,
    f.excess_quantity,
    f.snapshot_date
FROM fact_inventory_status f
JOIN dim_material m ON f.material_id = m.material_id
WHERE f.is_recoverable_asset = 1;
GO
```

Gold Tier Semantic Layer

```
CREATE OR ALTER VIEW dbo.vw_q2_mc_provisions AS
SELECT
    f.material_id,
    m.material_name,
    m.category,
    f.quantity_on_hand,
    (f.quantity_on_hand * m.unit_cost) AS inventory_value,
    f.snapshot_date,
    -- FIX: Healthy stock gets 0 provision. Bad stock gets capped at its true value.
    CASE
        WHEN f.is_mc_expired = 0 AND f.is_mc_dead_stock = 0 THEN 0
        WHEN f.mc_provision_amount > (f.quantity_on_hand * m.unit_cost) THEN (f.quantity_on_hand * m.unit_cost)
        ELSE f.mc_provision_amount
    END AS mc_provision_amount,
    CASE
        WHEN f.is_mc_expired = 1 THEN 'Expired'
        WHEN f.is_mc_dead_stock = 1 THEN 'Dead Stock'
        ELSE 'Healthy'
    END AS obsolescence_type
FROM fact_inventory_status f
JOIN dim_material m ON f.material_id = m.material_id
WHERE f.is_mc_material = 1;
GO
```

vw_q2_mc_provisions:

Evaluates and categorizes high-priority inventory into business states ('Expired', 'Dead Stock', or 'Healthy') while applying a defensive programming cap to guarantee realistic financial liability tracking.

```
-- View 6: Q7 - Impacted Sales Orders & Customers (Remains stable)
CREATE OR ALTER VIEW vw_q7_impacted_orders AS
SELECT
    io.sales_order_id,
    io.customer_name,
    io.finished_product,
    io.material_id,
    m.material_name,
    io.required_delivery_date,
    io.quantity_ordered,
    io.stock_cover_days,
    io.lead_time_days
FROM OPENROWSET(
    BULK 'https://ppgsg.dfs.core.windows.net/silver/impacted\_sales\_orders/\*\*',
    FORMAT = 'PARQUET'
) AS io
JOIN dim_material m
    ON io.material_id COLLATE Database_Default = m.material_id;
GO
```

vw_q7_impacted_orders:

Utilizes serverless OPENROWSET performance to directly map backend inventory deficits to external customer sales orders, revealing immediate fulfillment bottlenecks.

1st Dashboard

KPI card that shows the count of materials currently falling below their reorder threshold.

Donut chart summarised that most of them recoverable assets and 20.46% of them are at stockout risk.

KPI card that shows total amount that is at risk due to materials that fall under Magna Carta (MC) rules

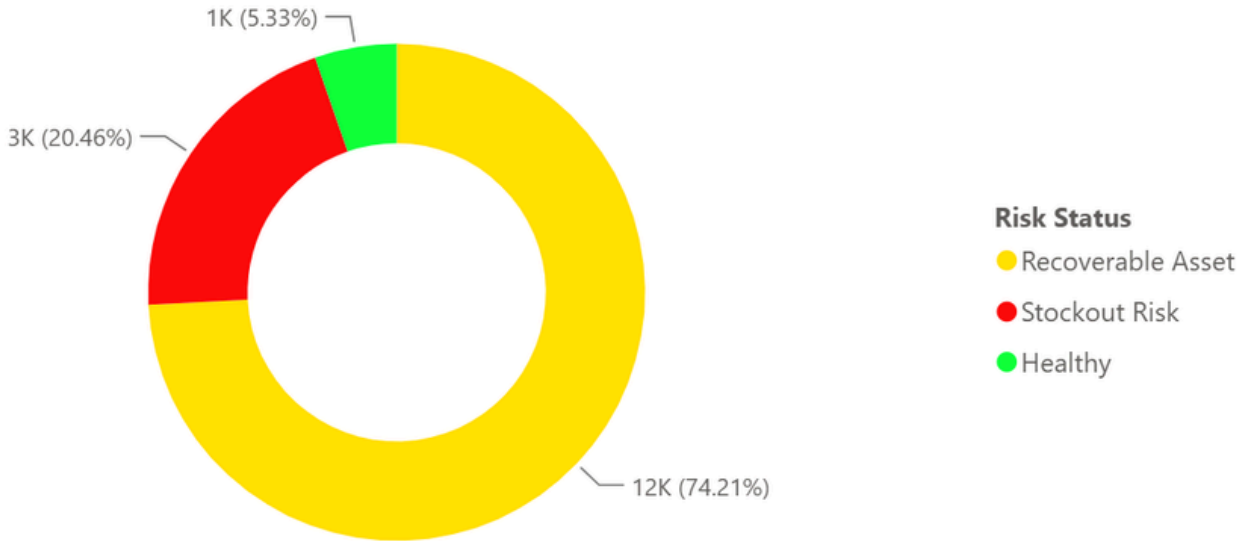
Total Financial Exposure



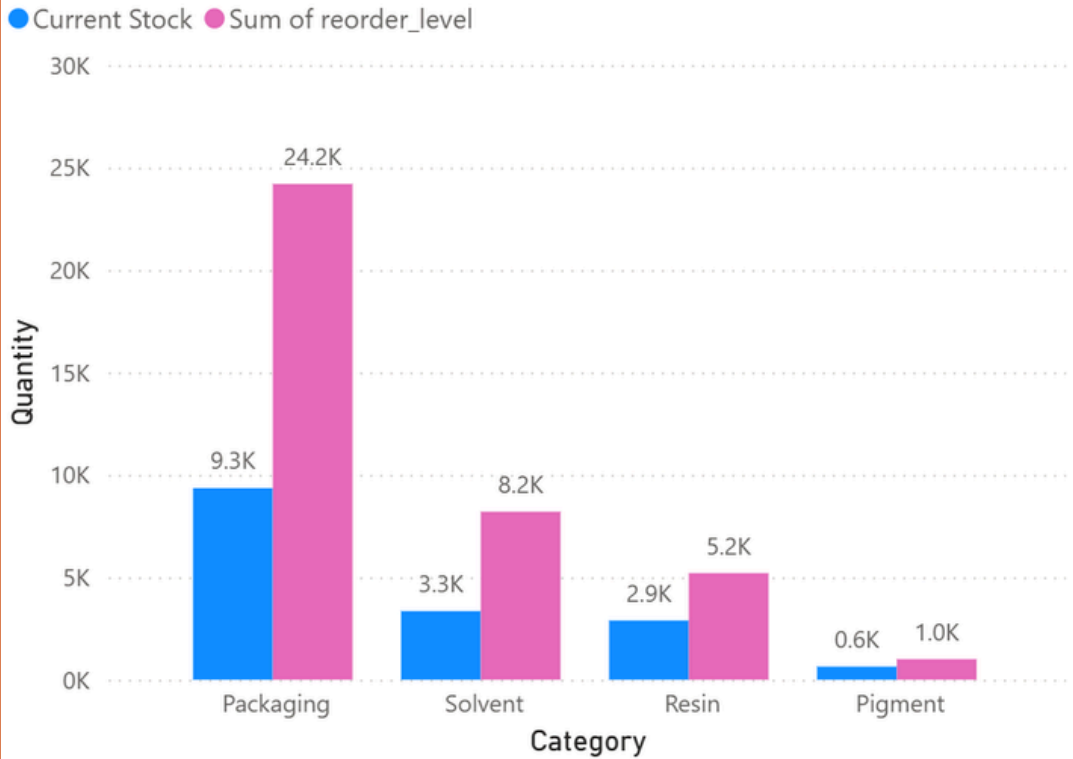
Critical Storage Count



Q2 Q6. Stockout risk vs Recoverable Assets



Q1. Current Stock vs Reorder Thresholds



material_id	material_name	category	Sum of quantity_on_hand	Risk Status
MAT001	Titanium Dioxide	Pigment	276	Stockout Risk
MAT002	Iron Oxide Red	Pigment	183	Stockout Risk
MAT003	Carbon Black	Pigment	179	Stockout Risk
MAT006	Alkyd Resin	Resin	381	Stockout Risk
MAT007	Acrylic Resin	Resin	347	Stockout Risk
MAT008	Epoxy Resin	Resin	104	Stockout Risk
MAT011	Xylene	Solvent	209	Stockout Risk
MAT013	Mineral Spirits	Solvent	210	Stockout Risk
MAT015	Butanol	Solvent	211	Stockout Risk
MAT016	Metal Can 1L	Packaging	349	Stockout Risk
MAT017	Metal Can 4L	Packaging	423	Stockout Risk
Total			16224	

A clustered bar chart that enables procurement teams to quickly identify which categories are most at risk of falling below safe stock levels and ready for ordering.

A material detail table that immediately identify which specific materials require urgent attention without needing to scan the entire dataset manually.

2nd Dashboard

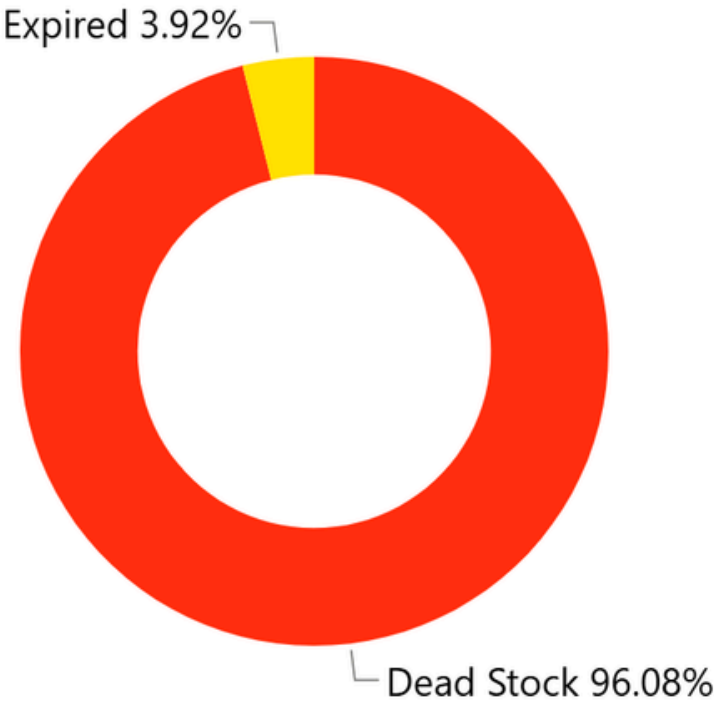
KPI card to track overall financial exposures.

Q4 Q5. Total MC Write-down Provision

RM3,516,990.87

Sum of mc_provision_amount

Q3. Financial Loss Split

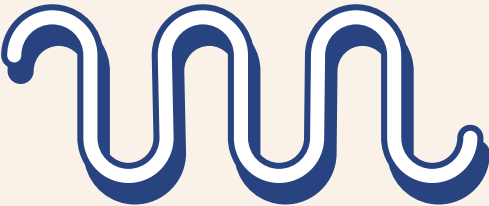


This donut chart indicates that the dominant financial risk comes from materials that have had no consumption for more than nine months (dead stock) rather than from materials that have passed their expiry date.

Outdated Assets at Risk

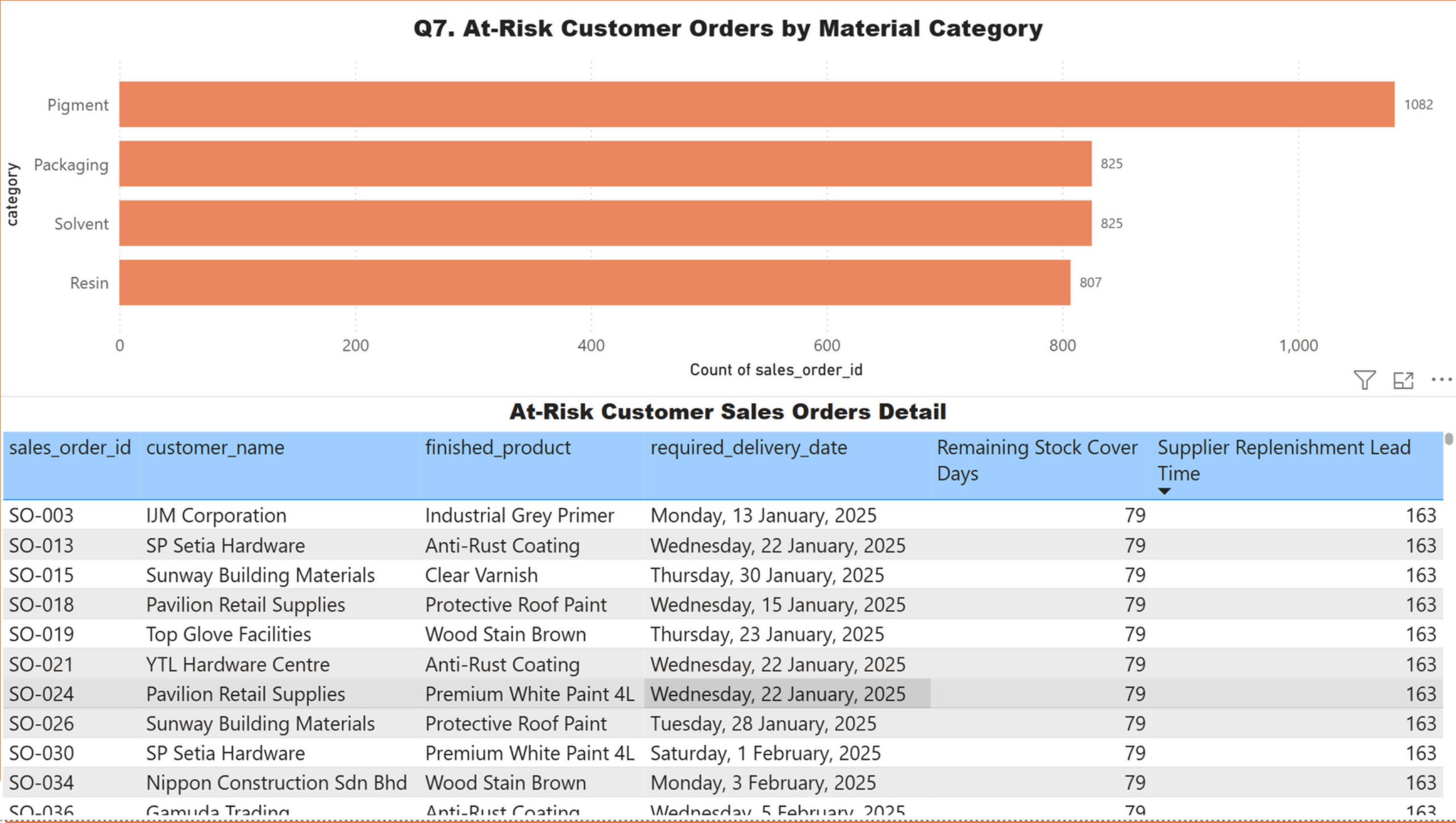
material_id	category	material_name	Sum of inventory_value	obsolescence_type
MAT053	Pigment	Premium Violet Pigment	342,491.97	Dead Stock
MAT052	Pigment	Pure Yellow Paste	308,176.00	Dead Stock
MAT055	Resin	Acrylic Copolymer	237,271.04	Dead Stock
MAT064	Packaging	Cardboard Box Medium	232,903.30	Dead Stock
MAT058	Resin	Epoxy Resin	190,198.50	Dead Stock
MAT069	Packaging	Cardboard Box 1L	169,020.02	Dead Stock
MAT061	Pigment	Premium Violet Oxide	167,824.58	Dead Stock
MAT074	Packaging	Plastic Bottle 500ml	149,581.25	Dead Stock
MAT066	Solvent	Ethanol (Standard)	146,089.02	Dead Stock
MAT054	Packaging	Cardboard Box 20L	145,282.68	Dead Stock
MAT057	Solvent	Butyl Acetate (Standard)	134,897.72	Dead Stock
MAT065	Solvent	Heptane (Grade A)	128,975.04	Dead Stock
MAT067	Packaging	Plastic Drum 500ml	126,705.39	Dead Stock
MAT070	Pigment	Organic Red Paste	120,104.79	Dead Stock
MAT072	Packaging	Steel Drum Small	117,745.43	Dead Stock
MAT051	Resin	Phenolic Hardener	116,974.08	Dead Stock
MAT073	Packaging	Plastic Drum Small	114,727.88	Dead Stock
Total			4,186,539.75	

A detailed table sorted in descending order by inventory value so that the highest-cost obsolete materials appear at the top of the list making it straightforward for the finance team to distinguish the worst write-down categories immediately.

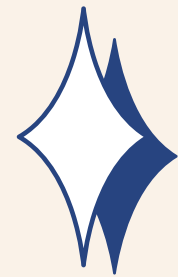


A horizontal bar chart is used and sorted in descending order to allow the supply chain team to prioritise their replenishment actions by category. Pigment is the most affected category with 1,082 impacted orders.

3rd Dashboard

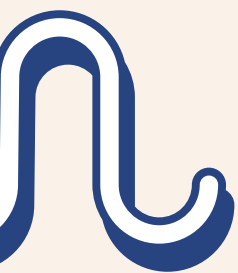


this table flags critical sales orders at risk of delivery failure because our current stock runs out in 79 days but new materials take 163 days to arrive.



Conclusion

- 01 In conclusion, this project successfully developed a data engineering solution to support PPG's supply chain and inventory risk management. By using the Medallion Architecture, raw inventory data was processed through the Bronze, Silver and Gold layers to produce clean, structured and analytics-ready data. The system also applied business logic to identify Recoverable Assets, Magna Carta Dead Stock, expired materials and stockout risks.
- 02 Through the Power BI dashboards, key inventory and financial risks can be monitored more clearly. The dashboards help highlight materials with high financial exposure, critical stock levels and customer orders that may be affected by stock shortages. Overall, the proposed solution improves decision-making by reducing manual analysis, increasing inventory visibility and helping the company respond earlier to financial and supply chain risks.



Thank You

